

Annexure-IV: MSDS of the formulation

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HINDUSTAN PETROLEUM GREEN R&D CENTRE,

KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, DEVANAGUNDI, HOSKOTE TALUK,
BANGALORE-560067.

PH: 080-2807-8607/2807-8520

MATERIAL SAFETY DATA SHEET

HP-SolarOKare

SECTION 1. CHEMICAL IDENTIFICATION:

NAME: HP-SolarOKare

Application: Solar panel cleaning liquid

Address:

Hindustan petroleum green R&D centre,

KIADB industrial area, Tarabahalli village, Devanagundi, Hoskote Taluk, Bangalore-560067.

Ph: 080-2807-8607/2807-8520

SECTION 2. COMPOSITION

Name	Content
Surfactants	0.30%
Mild acids	0.20%
Sodium benzoate	0.25%
Diluent Solvent (water)	98%

SECTION 3. HAZARDS IDENTIFICATION:

- Harmful by inhalation, in contact with skin and if swallowed
- The NFPA label for the formulation is given below.



SECTION 4. FIRST AID MEASURES:

- In case of direct contact with eyes or skin, immediately flush with copious amounts of water for at least 15 minutes. Also remove contaminated clothing and shoes.
- Ensure adequate flushing of the eyes
- If inhaled, take him/her to fresh air. If not breathing properly, give artificial respiration. If breathing is difficult, give oxygen.
- If swallowed, wash mouth with water provided the person is conscious. Call a physician.
- Wash contaminated clothing before reuse

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SECTION 5. FIRE FIGHTING MEASURES:

- Carbon dioxide
- Dry chemical powder
- Water spray

SECTION 6. ACCIDENTAL RELEASE MEASURES:

- Evacuate area
- Wear self-contained breathing apparatus, rubber boots and heavy rubber gloves
- Sweep up, place in a bag and hold for waste disposal
- Avoid raising dust
- Ventilate area and wash spill site with water after material pickup is complete

SECTION 7. HANDLING AND STORAGE:

- Wear appropriate NIOSH/MSHA – approved respirator, chemical – resistant gloves, safety goggles and protective clothing.
- Keep the chemical in tightly closed container. Store in a cool and dry place.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION:

- Wear appropriate NIOSH/MSHA - approved respirator, chemical-resistant gloves, safety goggles and protective clothing.
- Safety shower and eye bath
- Face shield
- Do not breathe dust. Do not get in eyes, on skin, on clothing.
- Avoid prolonged or repeated exposure
- Wash thoroughly with water after handling

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MATERIAL SAFETY DATA SHEET

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SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES:

Appearance: Colorless clear liquid

Density: 1000 – 1100 kg/m³ at 25 °C

Flash point: Not applicable

pH at room temperature: 6.5

SECTION 10. STABILITY AND REACTIVITY:

- Unstable if heated

SECTION 11. TOXICOLOGICAL INFORMATION:**ACUTE EFFECTS**

- Harmful if swallowed, inhaled, or absorbed through skin
- Symptoms of exposure may include burning sensation, coughing, wheezing, laryngitis, short ness of breath, headache, nausea and vomiting
- Inhalation may result in spasm, inflammation and edema of the larynx and bronchi, chemical pneumonitis and pulmonary edema

SECTION 12. ECOLOGICAL INFORMATIONS

- Data not yet available

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MATERIAL SAFETY DATA SHEET
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SECTION 13. DISPOSAL CONSIDERATIONS:

- Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber
- Observe all central (Govt. of India), State and local environmental regulations

SECTION 14. TRANSPORT INFORMATION

UN No : UN3082

Hazardous Class : Class 9- Non-Flammable liquids

HAZARDS IDENTIFICATION This chemical is not considered hazardous by the 2012 OSHA (Occupational Safety and Health Administration) Hazard Communication Standard (29 CFR 1910.1200)

Not a dangerous substance or mixture according to the Globally Harmon System (GHS)

HAZARDS Symbol

**SECTION 15. REGULATORY INFORMATION:**

- Harmful by inhalation, in contact with skin and if swallowed. In case of accident or if you feel unwell, seek medical advice immediately.
- In case of contact with eyes, rinse immediately with plenty of Water and seek medical advice
- Take off immediately all contaminated clothing
- Wear suitable protective clothing, gloves and eye/face

SECTION 16. OTHER INFORMATION:

The information and recommendations in literature and on labels are to the best of our knowledge reliable. However, nothing herein is to be construed as a warranty or representation. Users should make their own tests to determine the applicability of such information or the suitability of any products for their own particular purpose. License granted to make unlimited paper copies for internal use only.

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